

Project Proposal: Machine Learning with Graphs

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TL;DR. We represent a frozen ViT’s attention as a graph and train a GNN to predict failures beyond confidence and non-graph baselines. In completed follow-ups, adding graph-detector scores improves a LogitDynamics adaptation, while a matched set-based combination achieves almost identical observed performance.

1 Background & Motivation

Vision Transformers (ViTs) represent an image as patch tokens and repeatedly exchange information through multi-head self-attention [Dosovitskiy et al. 2021]. Reliability is usually estimated from the final logits, although predictions with similar confidence may arise from different layer-wise interactions. We therefore ask:

Can a GNN operating on a ViT’s multi-layer attention graph predict classification errors, especially under unseen corruptions, better than output-, representation-, and non-graph attention baselines?

Our completed follow-ups also examine complementary usefulness alongside a strong internal-feature predictor and the contribution of its feature groups.

This is useful for selective prediction, where likely failures are rejected or referred for review, and scientifically useful for testing whether errors exhibit recurring routing signatures. Each image naturally induces a directed, weighted graph over patch tokens, while correctness is a graph-level property of the frozen classifier.

Related work and gap. ViT attention has already been interpreted as a graph, but mainly for other objectives: relevance propagation for explanation [Chefer et al. 2021], weighted-PageRank token pruning [Wang et al. 2024], and graph regularization of class-patch interactions for weakly supervised segmentation [Yang et al. 2025]. Cultrera et al. [2023] use a convolutional autoencoder over ViT attention heatmaps for OOD detection, without learning on token-to-token connectivity. Closest in task, LogitDynamics predicts ViT errors from layer-wise class-logit trajectories, but does not use attention graphs [Beigelman and Freiman 2026].

NLP provides the closest methodological precedents. Handcrafted topological or spectral summaries of attention graphs have been used for generated-text and hallucination detection [Binkowski et al. 2025; Kushnareva et al. 2021], while HalluZig models how attention-graph topology evolves across layers [Samaga et al. 2026]. More directly, HaluGNN uses hidden states as node features and attention values as weighted edges [Kong et al. 2026b]. CHARM represents raw attention scores and activations as attributes of a token graph and applies edge-aware message passing; its ablations show a benefit from graph connectivity over a set model [Frasca et al. 2026]. CausalGaze further refines attention edges using gradient sensitivity before GNN classification [Kong et al. 2026a]. Thus, learned attention-graph diagnostics are established for language models. However, our literature search found no analogous method trained on per-image, multi-layer ViT patch-attention graphs to predict visual classification failure. We investigate this vision-specific gap.

2 Proposed Approach

This section describes the original proposal. The completed follow-ups in Section 4 use separately specified layer-specific detectors and score combinations; they do not implement the shared-GNN sequence model or establish the unseen-corruption objective of the original experimental plan.

Let $A^{(\ell,h)} \in \mathbb{R}^{N \times N}$ be the attention matrix at ViT layer ℓ and head h , including the class token. Since query token i reads information from token j , we orient information flow as $j \rightarrow i$. Following CHARM [Frasca et al. 2026], the raw per-head attention scores are used directly as edge features:

$$\mathbf{e}_{ji}^{(\ell)} = \left[A_{ij}^{(\ell,1)}, \dots, A_{ij}^{(\ell,H)} \right].$$

To obtain a sparse graph without discarding interactions that are strong in only one specialized head, we retain the edge $j \rightarrow i$ whenever

$$\max_h A_{ij}^{(\ell,h)} > \tau.$$

Thus, the threshold determines only whether the edge is present; once retained, its edge feature contains the original attention scores from all heads. The threshold τ is selected on validation data.

Diagonal scores are stored as node features, together with normalized patch coordinates and a class-token indicator. We compare this attention-focused representation with a richer variant that also appends the corresponding ViT token embedding $\mathbf{z}_i^{(\ell)}$.

An edge-aware message-passing GNN uses $\mathbf{e}_{ji}^{(\ell)}$ when computing messages. We construct one attention graph for every ViT layer,

$$G^{(1)}(x), \dots, G^{(L)}(x),$$

and encode all layer graphs with a shared GNN:

$$\mathbf{g}_\ell = \text{GNN}\left(G^{(\ell)}(x)\right).$$

The resulting ordered sequence of graph embeddings

$$\mathbf{g}_1, \dots, \mathbf{g}_L$$

is then combined and followed by an MLP that predicts classification failure. Using all layers allows the detector to capture the complete evolution of attention routing, including early local interactions and later global aggregation. As ablations, we will compare this full layer-wise model with a last-layer-only model and a final-four-layers model.

For image x with label $y(x)$ and frozen ViT f , the automatic target is

$$y_{\text{err}}(x) = 1 \left[\arg \max_c f_c(x) \neq y(x) \right].$$

In this proposed graph-detector setup, only the detector is trained, using class-weighted binary cross-entropy. We compare:

- (1) the attention graph with minimal node features;
- (2) the attention graph augmented with token embeddings.

Novelty. The novelty is not the attention-as-graph view or GNNs over Transformer traces in general. It is their controlled evaluation for per-image ViT patch graphs, with comparisons against output signals, hidden representations, and non-graph models given the same attention data.

3 Experimental Plan

Datasets and setup. The primary benchmark will use CIFAR-100 and CIFAR-100-C, whose corruption families and severity levels follow Hendrycks and Dietterich [2019]. We will begin with a publicly available ViT checkpoint fine-tuned on CIFAR-100 and verify its accuracy, preprocessing pipeline, label mapping, confidence distribution, and access to attention matrices from all Transformer layers. If no suitable checkpoint is available, we will fine-tune an ImageNet-pretrained ViT on CIFAR-100. The selected ViT will then be frozen throughout all detector experiments.

If no publicly available checkpoint satisfies these requirements, we will fine-tune an ImageNet-pretrained ViT on the clean CIFAR-100 training set and select the checkpoint according to validation accuracy. In either case, the selected ViT will then be frozen and used as the fixed classifier throughout all detector experiments.

We will evaluate the selected classifier on CIFAR-100-C and report accuracy separately for each corruption type and severity level. After validating the classifier, we will precompute its attention graphs. The detector will be trained on clean images and a selected subset of noise, blur, weather, and digital corruptions, while complete corruption families will be reserved for testing. All corrupted versions of the same source image will remain in the same split to prevent leakage. Optional far-OOD datasets are SVHN [Netzer et al. 2011] and DTD [Cimpoi et al. 2014]; localized patches and occlusions will be generated automatically.

Baselines and ablations. We first report the frozen ViT’s clean and corruption-specific classification accuracy. For failure prediction, we compare against maximum softmax probability, logit margin, and an MLP on the final class-token embedding.

To isolate the contribution of graph structure, we also compare against an MLP receiving the same flattened attention tensors as the GNN and a sequence model receiving class-token embeddings from all ViT layers.

Graph ablations include last-layer-only input, the final four layers, all layers, and threshold sensitivity.

Metrics and success criteria. We evaluate error prediction using AUROC and AUPRC, treating an incorrect ViT prediction as the positive class. Since the detector may be used to reject unreliable predictions, we also report risk-coverage curves, area under the risk-coverage curve, and selective accuracy at fixed coverage levels.

Metrics are reported separately for clean images, corruption families seen during detector training, and unseen corruption families. The main hypothesis is supported if the GNN outperforms non-graph models given the same attention values, particularly on unseen corruptions.

Compute resources. The project requires GPU access provided by the course or university. The main compute cost is fine-tuning the ViT and precomputing attention graphs, token features, and logits for clean and corrupted images, as well as training the GNN over them.

4 Controlled Follow-up Experiments

We completed three related comparisons: standalone graph/set detectors against a published alternative, score fusion with that alternative, and a decomposition of its features. Earlier group experiments already included internal-plus-output fusion and competitive rich non-graph controls; these follow-ups extend those questions rather than introduce fusion or set controls for the first time.

Implemented detectors. The preserved graph and set detectors use attention layers 3, 6, 9 and 12, with separate models for each layer and seed. Every graph/set arm receives the same H12 token representations within 784-dimensional node

features and 12 raw per-head attention edge features (the *raw12* regime). Per-head values at or below the fixed 0.02 sparsification threshold are zero-filled. The graph model uses two width-64 edge-gated mean message-passing layers, with source/target-conditioned gates and a CLS-plus-gated-node-sum readout. The width-84 set model separately encodes nodes and edges with two-layer MLPs. Its readout combines CLS, a gated node sum, node mean/max, edge mean/max and $\log(1 + \text{edge count})$. It keeps positions, CLS flags, diagonal attention and H12 values, but discards edge incidence and thus edge-to-endpoint association. Pooling also differs; matching feature values does not isolate message passing alone. G and S denote equal means of the four within-seed sigmoid detector scores, not shared-GNN sequences, union graphs or cross-seed prediction averages. The preserved fits used 20 epochs, AdamW, learning rate 0.002, weight decay 0.0001, dropout 0.15 and batch size 24; checkpoints maximized selection-set AUROC. Subsequent comparisons reused these fits, including existing GNN models, and are not independent training replications.

Our LogitDynamics adaptation, D [Beigelman and Freiman 2026], keeps the ViT-Base frozen and fits twelve auxiliary linear class heads on CLS states. Each head and the original classifier contribute six numeric features: the original predicted class's logit and the five largest competing-class logits. These 78 values and seven dynamics features feed a linear error detector. This is distinct from our earlier local CLS-trajectory GRU and is not a reproduction of the published ViT-Large experiments. Auxiliary heads use unweighted multiclass training for 16 epochs; error probes use weighted binary cross-entropy for 100 epochs, selecting the highest validation AP and the earliest exact tie. Inputs, supervision allocation and budgets differ from the graph/set pipelines.

Data roles and fusion. We use CIFAR-100 photographs with nine views each: clean, and Gaussian noise, motion blur, fog and JPEG at severities 3 and 5. All views of a photograph remain grouped. Graph/set fitting uses 1,600 source photographs and checkpoint selection uses 400. LD auxiliary heads use 1,200 sources; the error probe uses 800, and selection uses 400. Standalone evaluation uses the same 800 development photographs (7,200 examples) for all methods.

For fusion, a fixed outcome-independent hash partition divides these previously examined development photographs into 400 for fitting and 400 for assessment. The earlier 400-photo combiner pool was unsuitable here because LD used it for probe training. We fit separate two-score logistic combinations DG , DS and DD' per seed, with training-only standardization and fixed L2 regularization ($C = 1$; intercept; no class balancing). Rankings use decision logits, and D remains unrecalibrated. D' is another existing LD seed, paired as $7 \rightarrow 17$, $17 \rightarrow 27$, $27 \rightarrow 7$; these overlapping pairs are a generic ensemble control, not a compute-matched control. This linear score fusion is not the earlier conditional gate. Decomposition retains LD's original 800/400 training/selection roles and evaluates on all 800 development photographs. The studies are exploratory: separating combiner fitting and assessment does not create an untouched test set.

Statistical scope. AUROC measures error ranking, with incorrect ViT classifications positive. Tables report means of seed-specific metrics for seeds 7, 17 and 27, not metrics of cross-seed averaged predictions. AP denotes scikit-learn average precision, not trapezoidal PR area. The follow-ups use 2,000 paired source-photo bootstrap draws. Fusion separately resamples fitting and assessment photographs and refits all combiners; decomposition resamples evaluation photographs with fitted models fixed. The three primary contrasts $DG - D$, $DG - DS$ and $C - B$ use $98\frac{1}{3}\%$ Bonferroni-adjusted percentile intervals, targeting nominal 95% familywise coverage. Linear percentile endpoints are 0.0083333333 and 0.9916666667; only about 17 draws occupy each tail. Secondary intervals are descriptive 95% intervals. Uncertainty conditions on preserved base models and designated splits, includes combiner-fitting resampling where applicable, and does not cover all training variability or previous development-data exposure.

Earlier matched-control context. The separate September 11 *rich36* study found graph minus set AUROC $+0.006233$ (reported 95% interval $[0.002618, 0.009836]$, five seeds). Its attention-only counterpart gave -0.003686 , with interval $[-0.008349, 0.001045]$. Feature regimes, cohorts and training budgets differ from the later raw12 ensembles, so these are not contradictory controlled replications. An earlier two-seed replication found that a learned four-layer score stack improved on its learned single-final-layer control. No matched four-model final-layer ensemble was evaluated, so this does not isolate layer diversity or show that ensembling caused the change between studies.

Table 1. Separate standalone and fusion evaluations. Mean AUROC across three seeds; values should be compared only within a panel.

Standalone: 800 development photographs	
Method	Mean AUROC
MSP	0.861943
Full-logit MLP	0.875708
Set ensemble (<i>S</i>)	0.888559
GNN ensemble (<i>G</i>)	0.888895
LogitDynamics adaptation (<i>D</i>)	0.898066
Fusion: 400 assessment photographs	
Method	Mean AUROC
<i>D</i>	0.900126
<i>G</i>	0.895609
<i>S</i>	0.894954
<i>DG</i>	0.907329
<i>DS</i>	0.907286
<i>DD'</i>	0.902116

Standalone and complementary usefulness. On 800 development photographs, the prespecified $G - D$ difference was -0.009171 , with ordinary 95% interval $[-0.015874, -0.002361]$; *D* had higher AUROC in all three seed pairs (Table 1). This earlier comparison has its own single-contrast interval, without retroactive follow-up multiplicity correction. It establishes an advantage for the fitted LD adaptation over the preserved graph ensemble here, not over every group model, such as the separate rich conditional-gate pipeline.

On the distinct 400-photo fusion assessment, $DG - D = +0.007203$, adjusted interval $[0.003415, 0.011330]$. Although *G* underperforms *D* alone, its score improves ranking when combined with *D*. However, $DG - DS = +0.000043$, adjusted interval $[-0.001311, 0.001508]$; the set combination has almost identical observed performance, with no demonstrated graph-specific fusion advantage. This is not equivalence. The secondary $DG - DD'$ contrast is $+0.005213$, descriptive 95% interval $[0.001942, 0.008673]$. DD' also ranks above *D* by point estimate; this ordering is descriptive only.

Table 2. LD feature decomposition on all 800 development photographs. Only A/B/C readouts are newly fitted; *D* is the preserved full model.

Variant	Features	Mean AUROC
<i>A</i>	Six original-classifier features	0.863598
<i>B</i>	<i>A</i> plus the layer-12 auxiliary-head block	0.894831
<i>C</i>	Numeric blocks from all auxiliary layers and the original classifier	0.898966
<i>D</i>	<i>C</i> plus seven dynamics features	0.898066

Feature-group decomposition. With frozen auxiliary heads and training-only normalization, A/B/C use the original linear-probe recipe. Their newly generated cache reproduces preserved D predictions before fitting. The largest observed increment in this nested comparison is the final-layer auxiliary block: $B - A = +0.031233$, descriptive 95% interval $[0.025273, 0.037483]$ (Table 2). A is a six-feature readout, not the full-logit MLP. This contrast does not isolate auxiliary supervision from changes in projection, inputs and readout dimensionality.

Intermediate-layer projections provide a smaller additional benefit. The difference was $C - B = +0.004135$; its adjusted interval was $[0.000428, 0.007911]$. The small positive lower endpoint warrants restraint given the limited bootstrap tail resolution. This tests usefulness under the fixed recipe, not information absent from H12 or a unique explanation of the LD-GNN difference. Finally, adding the dynamics block slightly lowers AUROC: $D - C = -0.000900$, descriptive 95% interval $[-0.001750, -0.000048]$. The effect is small and close to zero; it does not establish that dynamics is generally harmful or unnecessary. The block includes class identities, not only summaries already present in C , so this is not a clean dynamics-versus-information-content test.

Interpretation and boundaries. These follow-ups distinguish standalone quality from complementary usefulness. Graph scores improve a stronger LD score, but a matched non-message-passing combination performs almost identically. This supports complementary detector outputs without establishing graph necessity. Attention-derived feature content, endpoint associations and graph processing are different questions; absence of message passing does not imply absence of attention information. The ablation identifies useful feature groups under a fixed readout, not a complete causal explanation. Score-level complementarity does not establish information absent from all LD features or final logits. Better error ranking neither improves the frozen classifier’s accuracy nor demonstrates calibration, threshold transfer or deployment reliability. These follow-ups do not validate fusion on unseen corruption families, and their near-tie does not invalidate the earlier positive rich-feature result.

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